Teacher Wu Kuan | Real Exam #3 — Taiwan GSAT Math A (2025) Q16: Why m = 0 Is Rejected

English lecture notes · YouTube Short LdVqtXP7BQ8

1. The trap question

L1 and L2 pass through (3, 1) with slopes m and −m. A circle centered at O is at distance 1 from L1 and tangent to L2. Find chord AB.

2. The method

- Distance from O to L1: $|1 - 3m| / \sqrt{(m^2 + 1)} = 1 \rightarrow m = 0$ or 3/4.
- m = 0 makes L1 = L2; the circle would be tangent, not cut a chord → reject. m = 3/4.
- L2: 3x + 4y − 13 = 0, radius r = 13/5.
- Chord = $2\sqrt{(r^2 - d^2)} = 2\sqrt{(169/25 - 1)} = 24/5$.

3. Parent check method

Ask your child: “After solving, does each answer still describe the picture in the problem?”

Practice

1. Find the distance from (0, 0) to the line 3x + 4y − 10 = 0.
2. The line 3x + 4y = 15 cuts the circle $x^2 + y^2 = 25$. Find the chord length.
3. A line through (2, 0) with slope m is tangent to $x^2 + y^2 = 1$. Find m.
4. The line y − 1 = m(x − 3) is at distance 1 from the origin. Find m.
5. The line x + y = 2 cuts the circle $x^2 + y^2 = 10$. Find the chord length.

Answers

1. 2
2. 8 (distance 3; $2\sqrt{(25 - 9)}$)
3. $m = \pm\sqrt{3}/3$
4. m = 0 or 3/4
5. $4\sqrt{2}$ (distance $\sqrt{2}$; $2\sqrt{8}$)

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